EVIDENCE SYNTHESIS AND CONCEPTUAL ANALYSIS

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Clinical–Laboratory Variability in Digitally Fabricated Complete Dentures:

A Critical Evidence Synthesis and Traceability Framework

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ABSTRACT **Purpose.** To evaluate how clinical records, laboratory fabrication and postprocessing contribute to variation in complete denture performance, and to define an outcome-linked framework for documentation. **Methods.** A critical narrative synthesis was developed from a curated set of 46 research, review, technique and methodological publications available through the review cutoff. Published effect estimates were retained with their original comparators and denominators; no new pooled estimate or systematic eligibility screening was undertaken. **Findings.** The principal procedural review incorporated 39 studies, while a network meta-analysis of 43 in-vitro studies favored milled bases for trueness and surface roughness under the studied methods. In a 60-base experiment, one printed resin had the lowest absolute deviation at both measured time points, whereas milled bases displayed the least change between 24 hours and six months. A 2026 synthesis of 25 randomized trials found no clear superiority for patient-important endpoints and identified only a small, low-certainty retention difference favoring printing over milling. Scanning, design transfer, curing, bonding, inspection and follow-up represent different potential sources of variability. **Conclusions.** Manufacturing route alone is an inadequate proxy for clinical quality. A six-checkpoint traceability model is proposed to link verified records and measured process deviations with maintenance and patient-reported outcomes. The model is an analytical proposal and requires prospective evaluation.

CLINICAL RELEVANCE A base can be dimensionally accurate yet fail to meet a patient's functional requirements. Documentation of the clinical input, production route, quality checks and subsequent adjustments allows this distinction to be investigated.

KEYWORDS Complete denture; CAD–CAM; prosthodontics; 3D printing; milling; dimensional accuracy; dental laboratory; patient-reported outcomes.

1. Introduction

The fabrication of a complete removable denture requires more than the accurate reproduction of a digital design. The clinical impression must capture relevant supporting and peripheral tissues; jaw relations, tooth arrangement, vertical dimension and esthetic information must be transferred; the base and denture teeth must be produced and joined; and the finished appliance must be evaluated in use. Each stage is a possible source of departure from the intended treatment. An apparent advantage in a bench test does not necessarily translate into patient satisfaction or fewer corrective visits.[1, 2, 3]

A systematic review by Thu et al. described marked diversity in clinical and laboratory procedures across 39 studies of digital complete dentures.[1] Subsequent research has generally approached isolated components of the process: base trueness and roughness, mechanical performance, interfaces, microbial behavior, occlusion and patient outcomes.[4, 5, 6, 7, 8, 9, 10] Other syntheses evaluate clinical comparisons or umbrella-level outcomes; their primary studies may overlap.[11, 12, 13, 14, 15, 16, 17, 18, 19, 20]

The clinical question is therefore not simply whether printing, milling or conventional processing is preferable. It is *where* clinically relevant variation arises, *which* outcomes reveal its consequences, and *how* evidence can connect the work performed in the dental laboratory with the patient's subsequent experience. This review synthesizes the available research across those questions and articulates a testable documentation framework.

2. Methods and evidence appraisal

2.1 Review design and identification

This work is a **critical narrative review**, not a systematic review, a new meta-analysis, or an experimental validation study. The evidence corpus contains 46 referenced publications available through the review cutoff. Studies were identified through targeted searches in PubMed, access to journal and publisher records, and examination of relevant

review reference lists. Searches combined concepts equivalent to *complete denture*, *edentulous arch*, *CAD/CAM*, *milled denture*, *3D-printed denture*, *laboratory workflow*, *accuracy*, *postcuring*, *bond strength*, *retention*, and *quality of life*. Bibliographic identity and the main reported estimates used in the text were checked against accessible article records and full texts where available.

The sources were chosen for relevance to four prespecified interpretive domains: clinical-record validity; geometric accuracy and stability; material, surface and interface behavior; and patient-level outcomes. Clinical studies, in-vitro experiments, systematic reviews/meta-analyses, dental-technique reports and a narrative-review appraisal paper were eligible for the conceptual synthesis. Implant-retained overdentures and fixed prostheses were not treated as evidence about tissue-supported complete dentures. No quantitative threshold for inclusion based on effect size or significance was applied.

This was a purposive rather than exhaustive selection. A saved database-by-database search export, formal screening flow, protocol registration, duplicate independent selection and study-level risk-of-bias assessment are not available for the present synthesis. Consequently, *46 denotes cited sources, not 46 independently eligible trials or a complete census of the literature*. SANRA criteria informed the emphasis on a clear question, accessible sources and the distinction between evidence and interpretation.[21]

2.2 Rules for interpreting the literature

Results were read at the level at which they were measured. The denominators of specimens, denture bases, complete appliances and patients were not interchanged. Laboratory deviation (μm), flexural strength (MPa), retention force (N), microbial assay readouts, time, complications and oral health-related quality of life (OHRQoL) were considered distinct endpoints. Published meta-analytic estimates were reproduced only with their specific comparator and statistical coding. A non-significant difference was not interpreted as equivalence unless an appropriate equivalence or non-inferiority analysis was reported.

Reviews addressing similar questions were interpreted as *overlapping evidence maps*, not independent replications. Full-text details were used when obtainable; otherwise claims were limited to abstract-level or publisher-record information. There was no reanalysis of individual patient data or raw laboratory datasets.

3. Clinical records before manufacture

3.1 Functional anatomy remains a limiting input

A precisely milled or printed base cannot correct an inadequate registration of movable mucosa or an unstable jaw-relation record. In Thu et al., 24 of 39 included studies used border molding and only 13 reported a trial denture stage. One contributing comparison recorded maxillary-base retention of 14.5–16.1 N after border-molded impressions versus 6.2–6.6 N after intraoral scanning; the figures represent *one study*, not pooled estimates.[1] Chebib et al. investigated fit and retention in a clinical crossover comparison of conventional impressions and digital scans.[22]

Two later reviews make the boundary conditions clearer. Mohamedin et al. assessed 26 reports and found insufficient evidence that digital scans could entirely replace conventional impressions in complete denture treatment, particularly where soft tissue mobility and vestibular anatomy affect the record.[23] Nguyen et al. synthesized 31 reports of functional impressions within digital workflows and similarly emphasized dependence on hybrid procedures for recording dynamic border morphology.[24] Neither result establishes that scans are clinically unusable; rather, digital capture cannot be treated as equivalent to a functional impression for all edentulous anatomies.

3.2 The design should be traceable to its clinical origin

The transfer from impressions or scans to digitized casts, jaw records and tooth arrangement is as consequential as printer or milling-machine precision. Technical papers describe combined analog–digital approaches, facially guided design, improved tooth positioning and dynamic occlusion registration.[25, 26, 27, 2] These reports demonstrate feasible procedures; they do not by themselves establish comparative patient benefit or reduced remake rates.

4. Dimensional accuracy and stability

4.1 What the laboratory measurements support

A network meta-analysis by Menon et al. included 43 in-vitro studies of denture bases. Milled bases showed higher trueness than compression-molded bases (source standardized mean difference [SMD] -2.25; 95% confidence interval [CI] $-4.09, -0.40$) and printed bases (SMD -1.57; 95% CI $-3.14, -0.01$); surface roughness comparisons also generally favored milling.[5] These are comparisons of laboratory endpoints, not estimates of clinical retention, comfort, repair rates or appliance survival.

Experimental findings are not uniform at the product level. Graf et al. compared manufacturing routes including milling and several printing systems and reported changes along different anatomical axes; subsequent handling and support re-

moval affected the measured result.[28] Cameron et al. compared the intaglio accuracy of seven fabrication systems, illustrating variation between implementations of the same broad manufacturing category.[29]

Casucci et al. evaluated 60 bases (10 milled, 10 conventional and 40 printed). At 24 hours, the Temp Print printed group had the lowest reported root-mean-square (RMS) deviation (54.20 ± 9.44 μm). It also had the lowest absolute RMS values at six months, *but the milled group showed the least change between time points*.[30] Thus, lowest measured deviation at one or more time points and smallest change over time are separate properties. The same experiment reported material-specific changes, including a six-month RMS of 194.1 ± 49.63 μm for Denture 3D+. One manufacturer's experimental result should not be presented as the universal superiority of printed bases.

4.2 Manufacture is not the end of dimensional change

Polymerization, cleaning, storage and thermal cycling can alter the geometry measured at release or follow-up. Zeidan and Helal investigated trueness and precision after thermocycling, while Casucci et al. examined a six-month interval.[31, 30] The measurement occasion and the area registered are therefore integral to the result: a whole-surface RMS value can conceal local flange distortion or a clinically important border defect. A valid laboratory report should state the reference surface, registration algorithm, inspected regions, timing, conditioning protocol and decision rule.

5. Material behavior, postprocessing and interfaces

5.1 Printing and curing are experimental variables

Material and machine labels are inadequate descriptions of a laboratory treatment. The build axis, nesting and support geometry, cleaning chemistry and duration, light exposure, temperature and aging regimen can modify the outcome. Jafarpour et al. found higher flexural strength at 90° than at 0° and 45° in one DLP resin and system (73.7, 55.2 and 61.8 MPa, respectively), whereas microhardness, elastic modulus and wettability followed different patterns.[32] This result should not be generalized as a universally optimal orientation; another resin and protocol may behave differently. Dzhondrova et al. and Aati et al. likewise examined effects of orientation or light-curing duration.[34, 35]

Fouda et al. investigated 180 specimens of two printable resins under different postpolymerization durations and thermal cycling conditions, addressing strength, accuracy and translucency rather than a single composite performance score.[36] Additional in-vitro studies quantify mechanical and surface properties of photopolymers versus milled or heat-polymerized materials.[37, 38, 39, 40, 41] Syntheses by Menon et al. and Azab et al. show that aggregate material performance varies across testing standards and compositions.[6, 7] Flexural strength measured in a uniform bar cannot be converted directly into a patient's probability of a flange fracture or denture-tooth separation.

5.2 Interfaces and microbiological measurements

The denture-base polymer is not the only material boundary. Tooth bonding, relining, repair surfaces and polished tissue-contacting regions require separate evaluation. A systematic review by Arzani et al. included 20 in-vitro studies and 156 comparisons. For tooth bonding, the initial overall analysis was nonsignificant; a sensitivity analysis excluding two outlier comparisons favored milling (mean difference -2.43 MPa, 95% CI $-3.90, -0.96$, under the original contrast). Reline bonding favored milling, with SMD -2.62 (95% CI $-3.22, -2.03$).[8] This distinction between initial and sensitivity analyses is essential for fair reporting.

A separate review by the same research group pooled 12 in-vitro studies and found higher microbial metabolic activity (SMD 1.285, 95% CI 0.810–1.760) and cell count (SMD 1.050, 95% CI 0.124–1.976) on printed compared with milled bases.[9] Such endpoints describe laboratory assays and do not establish an equivalent difference in clinical denture stomatitis. Any causal explanation must additionally address surface treatment, roughness, microtopography, cleaning exposure, aging, the test microorganism and intraoral ecological conditions.

6. Clinical outcomes and comparator integrity

6.1 The newest trials do not support a single winner

Srinivasan et al. compared milled and printed removable complete dentures in a 15-participant crossover randomized trial, while Iwaki et al. compared printed and conventional dentures in 18 participants and reported noninferiority for the prespecified OHRQoL outcome.[42, 33] Clinical comparisons by Deng, Peroz and Abo Heikal and colleagues address different routes, protocols, outcomes and follow-up intervals.[43, 44, 45] Their findings should not be collapsed into a claim that *digital* dentures, treated as a homogeneous class, always outperform conventional appliances.

The comprehensive review by Prott et al., published 26 October 2026, included 25 randomized trials (27 reports). It found no clear advantage for any fabrication method on patient-important outcomes. The only statistically significant pooled comparison favored printed over milled dentures

Table 1: Selected quantitative findings with original denominators and comparisons. Values are reproduced from the cited studies; no effect was recalculated for this review.

Source	Evidence unit	Reported finding	Interpretation boundary
Thu et al.[1]	39-study systematic review	In one contributing comparison, border-molded bases 14.5–16.1 N versus scanned bases 6.2–6.6 N; 205–233 clinical min saved in two other studies.	Individual-study values, not pooled clinical effects.
Menon et al.[5]	43 in-vitro studies; network meta-analysis	Milled versus compression molded trueness: SMD -2.25 (95% CI $-4.09, -0.40$); milled versus printed: SMD -1.57 ($-3.14, -0.01$).	Laboratory trueness with source direction; not a measure of patient comfort.
Casucci et al.[30]	60 bases; three production routes	Temp Print: RMS 54.20±9.44 μm at 24 h; milled bases were most stable across six months.	Absolute fit and temporal change are different outcomes.
Jafarpour et al.[32]	One resin and DLP system; laboratory specimens	Flexural strength: 90°, 73.7 MPa; 45°, 61.8 MPa; 0°, 55.2 MPa.	Build orientation depends on resin, geometry and postprocessing.
Arzani et al.[8]	20 in-vitro studies; 156 comparisons	Reline bond strength favored milled bases, SMD -2.62 (95% CI $-3.22, -2.03$); source coding retained.	Bonding endpoint, not a clinical incidence of repair.
Arzani et al.[9]	12 in-vitro studies	Microbial metabolic activity: printed versus milled SMD 1.285 (95% CI 0.810–1.760).	Laboratory assay, not clinical stomatitis risk.
Iwaki et al.[33]	Randomized crossover trial; 18 participants	Printed dentures met the study's prespecified OHRQoL noninferiority criterion; satisfaction difference was not significant.	Not evidence for lifelong equivalence or all patient groups.
Prott et al.[18]	25 randomized trials; retention contrast from 2 trials ($n = 51$)	Printed versus milled retention: MD -1.80 N (95% CI $-2.49, -1.11$), favoring printing under source coding; low certainty.	Clinical importance uncertain; most other outcomes very-low certainty.

in retention force (mean difference -1.80 N, 95% CI $-2.49, -1.11$); the evidence was graded low certainty, with no established minimally important retention difference. Most other outcomes had very-low-certainty evidence.[18] A separate 2026 meta-analysis favoring conventional rather than printed dentures for retention ($P = .04$) addresses another comparison and does not logically contradict the printed–milled estimate.[17] Statistical nonsignificance is not proof of equivalence.

Importantly, Prott et al. found that in 13 of 21 within-participant trials, the study arms were fabricated from the same clinical records; six used separate records for the digital arm and two had mixed input structures.[18] This design feature helps separate the effect of the *fabrication route* from that of impression technique and jaw-relation recording. Where the same record is used for two methods, a favorable fabrication result cannot be generalized to a wholly digital impression workflow.

6.2 Aftercare and cost complete the measurement

Thu et al. found time savings of 205–233 clinical minutes in two contributing studies and reported study-specific maxima of 37.5% loss of retention and 31.3% remakes.[1] These maxima should not be interpreted as pooled complication rates or predictions for contemporary practice. A valid comparison of workflow efficiency should include chairside adjustments, unplanned visits, remake costs, technician time, material replacement and equipment amortization, rather than only the number of initial appointments. Reviews focused on costs and patient-reported outcomes reinforce the importance of separating these endpoints.[12, 15, 16, 11]

7. An outcome-linked traceability framework

7.1 Relationship to existing dental workflows

Fully or partly digital procedures already support design-file storage, reproducibility and long-term traceability. Grande et al. described three clinical steps and two laboratory phases, while Zhang et al. detailed a fully digital workflow integrating facial and intraoral records, interim verification and an archive supporting long-term digital traceability.[2, 46] The proposal here does **not** claim to be the first digitally traceable denture workflow. Its narrower contribution is a *research-oriented linkage* between record validity, production parameters, prespecified inspection regions, corrective actions and downstream patient-level outcomes. It addresses an information gap rather than introducing a new fabrication technology.

Table 2: Laboratory–Patient Traceability Framework (LPTF): proposed documentation fields and linked observations. These checkpoints are conceptual research variables, not validated performance scores.

Checkpoint	Minimum record required	Linked observation or corrective decision
1. Clinical input	Impression type; arch and supporting anatomy; functional border strategy; digital source revision; jaw-relation approval.	Clinician acceptance; documented tissue-border limitations; need to repeat impression or jaw record.
2. Design approval	CAD file/revision; occlusal scheme; vertical dimension; tooth arrangement and library; trial modifications; final approval.	Cause and frequency of occlusal and esthetic adjustments; design-related rework.
3. Manufacture	Route and system; resin/disk, batch and lot; machine; build axis or nesting; layer thickness or milling strategy; operator.	Whole-surface and region-specific deviation; detectable printing or milling defects.
4. Postprocessing	Cleaning agent/time; support removal; light-curing unit, duration and temperature; polishing; storage conditions.	Post-cure dimensional change, surface finish, bond and material-related corrective actions.
5. Release verification	Reference model; scan and registration method; predefined measurement regions; documented product-specific acceptance rule; reviewer.	Release, rework or remake decision recorded before patient fitting; reason for rejection.
6. Follow-up	Pseudonymized patient and prosthesis IDs; delivery date; standardized retention test; validated OHRQoL measure; aftercare visits; repairs/remakes.	Patient-level benefit and maintenance burden, linked to all previous checkpoints.

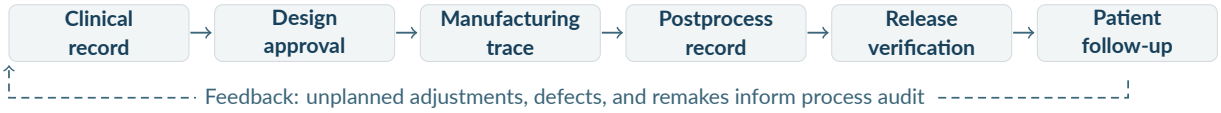


Figure 1: Conceptual flow of information across prosthesis manufacture and clinical use. Arrows denote planned record linkage and feedback, not empirically established causation or the validation of a new standard.

7.2 Six checkpoints and an explicit unit of analysis

The proposed Laboratory–Patient Traceability Framework (LPTF) comprises six linked checkpoints: (1) clinical-input validation; (2) approved design; (3) manufacture; (4) postprocessing; (5) release inspection; and (6) clinical follow-up. The primary observational unit for process analysis is a unique prosthesis, linked by a pseudonymous identifier to a patient and an arch. Separate denture bases, trial prostheses and remakes must not be counted as separate patients. Table 2 specifies minimal variables and their intended research linkage.

7.3 A validation design that would test the contribution

To determine whether such documentation improves care, a future multicenter protocol should select a prespecified patient-centered or service-level endpoint, such as unplanned definitive remake within 90 days, and compare periods or sites with and without implementation. Secondary outcomes would include unplanned visits, adjustment minutes, validated OHRQoL change, retention, repair burden, laboratory rejects and total cost per accepted prosthesis. Analyses would account for site, clinician, technician, material lot, anatomy, previous denture experience and multiple prostheses nested within the same patient.

Feasibility, missingness, interobserver agreement for release classifications and predictive validity would need to be established before effectiveness could be tested. The framework does not stipulate a universal allowable deviation in micrometres or force in newtons: such thresholds must be predefined for the particular clinical question and measurement system, then evaluated against patient-important outcomes.

8. Evidence overlap and causal boundaries

Review counts are not additive. Thu et al.'s 39 studies describe a broader procedural literature; Menon et al.'s 43 studies are in-vitro investigations; Prott et al.'s 25 trials assess clinical outcomes; and Ribeiro et al. synthesize already published reviews.[1, 5, 18, 15] A specimen included in a materials review might be discussed in another review of mechanics; the same patient cohort can be reported in multiple clinical papers. The present synthesis therefore uses these sources to clarify the *direction and limits* of findings, never to calculate a combined total number of independent experiments.

Table 3: Evidence-level distinctions relevant to interpretation; study collections may overlap and their counts must not be added.

Evidence source	Primary question represented	Inference not supported
Thu et al.[1] (39 studies)	Clinical records, fabrication steps and follow-up practices.	The causal contribution of each step to clinical success.
Menon et al.[5] (43 in-vitro studies)	Comparative laboratory trueness and surface properties.	Patient preference, long-term function or repair incidence.
Arzani et al.[8, 9] (20 bonding / 12 microbial studies)	Two distinct laboratory endpoints for printed and milled materials.	Direct estimates of clinical repair or denture stomatitis.
Prott et al.[18] (25 trials; 27 reports)	Comparative clinical and patient-reported outcomes across fabrication routes.	Whether a laboratory documentation protocol improves those outcomes.
Ribeiro et al.[15] (overview)	Summary of published reviews of clinical performance and costs.	Additional independent patients beyond those already included in the reviews.

The distinction also limits causal claims. An input-based trial that uses identical clinical records for milled and printed arms isolates a different question from a trial comparing an entire conventional workflow against a fully digital one. A mechanical performance gap in laboratory bars can guide process investigation but cannot identify a clinical safety threshold without prospective correlation. This hierarchy should shape both clinical interpretation and future audit design.

9. Limitations

The narrative corpus is purposive, not exhaustive, and study retrieval was not accompanied by a reproducible database search history or formal eligibility adjudication. Some published findings were reviewed from indexed abstracts or article-level summaries, not re-extracted individual-level data; formal risk-of-bias scores and a new systematic overlap matrix were not generated. Several reviews share research populations, and manufacturing methods and resin formulations evolve rapidly. Clinical trials are frequently small and often mix different baseline protocols and follow-up durations. These limitations restrict comparative certainty but do not invalidate the value of distinguishing evidence domains. Lastly, the LPTF has not yet undergone feasibility, reliability, discriminative-validity or intervention testing.

10. Conclusions

Across the published clinical and laboratory literature, complete denture quality depends on the validity of clinical records, design transfer, the chosen production and postprocessing procedures, inspection and subsequent clinical use. Milling performs well for selected dimensional and surface endpoints, yet some printed materials achieve favorable values under particular protocols. Absolute geometric deviation, stability over time, flexural properties and biological assays represent different measurements. The most recent trial-level synthesis has not demonstrated consistent superiority of any fabrication route for outcomes of greatest importance to patients. Accordingly, selection of a workflow should reflect anatomy, available records, manufacturing reliability, resource use and realistic follow-up requirements. Linking process deviations with documented corrective actions and patient outcomes provides a definable research strategy; its impact remains to be established prospectively.

Declarations

Funding: This work received no external funding. **Competing interests:** The author declares no competing interests. **Ethics:** No new participants, medical records or laboratory specimens were collected. **Data availability:** No original dataset was generated; data discussed are available in the cited publications. **Author contributions:** The author developed the manuscript's conceptual synthesis and reporting framework.

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